

Enhancement in the Spatial domain

Definition

* Image Enhancement is the process of improving the quality of an image by modifying its pixels directly

Gray Level Transformations

- * modify pixel intensity values
- * Eg. Negative, Log transformation
- * used to improve brightness

Histogram processing

- * Improve image contrast by distributing gray level
- * Tech: Histogram Equalization, Histogram stretching
- * Suitable for Low contrast image

Spatial Filtering

- * Smoothing filters remove noise
- * Sharpening filters enhance edges.

Application and Conclusion

- * used in medical imaging satellite imaging
- Surveillance, OCR and industrial inspection
- * spatial domain enhancement improves image quality. making images suitable.

Image Smoothing

Reduce noise and
transmission

Blurry, softer
edges.

Low pass
filtering

Integration or
summation

All positive
co-efficient.

Image Sharpening

Enhance contrast and
edges

Shp. defined
lines

High pass
filtering

Differentiation

min of positive
and negative

Application

Smoothing : medical images, CCTV. image processing

Sharpening : OCR : face recognition.

Industrial inspection

* Smoothing is used before image analysis to remove noise. while sharpening is used to highlight important features and edges

Image Enhancement in the frequency domain

Definition

* Frequency domain enhancement improve image quality by processing frequency components
Fourier transform

Low Pass filter (LPF)

- * Allows low frequency components to pass
- * Removes high frequency noise
- * Produces a smooth image

High Pass filter (HPF)

- * Allows high frequency components to pass
- * Enhances edges and fine details.
- * Produces a sharper image

Comparison

LPF: ~~noise~~ reduction and smoothing

HPF: Edge Enhancement and feature extraction

Application

LPF: medical imaging - image restoration

HPF: Object detection - fingerprint recognition

A.

Concept of image segmentation

Definition

Image segmentation divides an image into meaningful regions for easier analysis

Point Detection

- * Detect isolated pixels
- * used for fault or defect detection

Line and edge detection

- * Line detection identifies straight lines.
- * Edge detection identifies object boundaries using Sobel, Prewitt, or Canny operations

Thresholding and region based segmentation

* Thresholding. separates foreground and background using intensity values

* Region based groups neighbouring pixels with similar properties.

Application

- * medical image analysis
- * object recognition
- * Traffic sign detection

Importance of Edge Detection, Edge Linking

Edge Detection

- * Detects sudden changes in image intensity
- * Identifies object boundaries
- * method: Sobel, Prewitt, Canny

Edge Linking

- * Connects broken edge segment into continuous edge
- * Improve object shape detection

Boundary detection:

- * Detect the complete outline of an object
- * Helps measure object size and shape

Importance

- * Improves object recognition
- * Helps detect defects accurately
- * Essential in medical imaging and industrially

Applications

- * Face recognition, medical diagnosis, Fingerprint Recognition, quality inspection
- * Autonomous vehicle

Edge detection finds boundary edge linking connects them. boundary detection identifies complete object shapes.